

Preserving Evidence Anchors: Query-Rewriting Guardrails for Retrieval-Side Grounding

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Abstract

Query rewriting is common in retrieval-augmented generation, yet it can silently damage grounding before generation begins: a fluent rewrite may drop the exact terms that retrieve the evidence. We study this failure mode across five benchmarks, three retriever families, three rewriting strategies, and two open-weight 7B checkpoints. With Zephyr-7B, unconstrained rewriting degrades nDCG@10 by more than 0.05 for 26–76% of queries, and harmful cases often show *evidence-anchor drift*: rare terms, entities, and numerics replaced by weaker paraphrases. We propose a training-free layer that identifies candidate anchors, prompts the LLM to preserve them, verifies preservation deterministically, and falls back to the original query when verification fails. It needs no retriever training, logit access, or constrained decoder, and cuts mean qrel-observed harm from 48% to 10% on BEIR and from 26% to 6% on Natural Questions. Every guarded condition also applies identity fallback, and direct rewriting an edit-ratio cap, so this reflects the guardrail as a whole rather than anchor preservation alone. We measure retrieval-side evidence exposure, not generated-answer faithfulness. The principle is narrow but practical: LLMs may expand queries, but the retrieval interface should protect the anchors evidence access depends on.

1 Introduction

Grounded language-model systems depend on a brittle interface: in retrieval-grounded systems whose answers and citations are restricted to retrieved sources, the text sent to the retriever determines which evidence the generator can use. Modern RAG pipelines increasingly insert an LLM query rewriter at this interface, assuming that the rewrite will clarify the user’s intent or at worst behave like a no-op. This assumption is unsafe. A rewrite can remain semantically plausible while re-

moving the exact terms that retrieve the supporting documents. For example, “*Does ibogaine reduce Parkinson’s tremors?*” may become “*How do psychoactive compounds affect movement disorders?*”; the paraphrase is fluent, but it drops the anchors *ibogaine* and *Parkinson’s* that make the query evidence-seeking rather than generic.

We study query rewriting as a grounding interface rather than as a standalone retrieval optimization. A grounded generator can only cite, attribute, or reason over evidence that the retrieval stage exposes. Evidence lost during rewriting therefore creates an upstream failure mode: the generator may behave faithfully with respect to its retrieved context while that context no longer contains the evidence needed to answer the original question. Our claim is deliberately retrieval-side: we show that a rewriting component is less likely to remove benchmark-labeled evidence from the generator’s reachable context, not that a downstream generator will necessarily produce more faithful answers.

We call this failure *evidence-anchor drift*. It is a grounding failure that occurs before answer generation: if the retriever is sent a corrupted query, the downstream language model is asked to reason over the wrong evidence set. Existing work on LLM query expansion, including HyDE and Query2doc, emphasizes average retrieval gains (Gao et al., 2023a; Wang et al., 2023; Jagerman et al., 2023; Bigdeli et al., 2026b). Recent studies show that rewriting or expansion can hurt strong retrievers, ambiguous queries, or domain-specific queries (Weller et al., 2024; Killingback and Zamani, 2025; Abe et al., 2025). We extend this line from diagnosis to a lightweight systems remedy.

Our intervention separates what the LLM may add from what it may not remove. We identify a small set of candidate evidence anchors in the original query—rare terms, named entities, and numerics—and require the rewrite to preserve them. Compliance is checked by deterministic

validation; if validation fails, the system returns the original query. This *verify-then-fallback* design does not guarantee that every accepted rewrite helps, but it reduces the probability and severity of retrieval damage while keeping useful expansion available for underspecified queries.

Our contributions are:

- We measure the retrieval harm caused by unconstrained rewriting across five datasets, three retriever families, three rewriting methods, and two open-weight 7B checkpoints, and describe the evidence-anchor drift pattern we observe in harmful cases. Unconstrained rewriting harms 26–76% of queries under a $\Delta \text{nDCG@10} < -0.05$ threshold.
- We evaluate a deliberately minimal, training-free token-preservation layer for retrieval-risk-reducing rewriting. It requires no retriever retraining, no logit access, and no special decoder, making it applicable to black-box LLM-mediated rewriting when corpus IDF statistics are available.
- We show that the constraint layer reduces harm from 48% to 10% on BEIR and from 26% to 6% on Natural Questions, while keeping mean retrieval effectiveness near identity and allowing gains on underspecified queries.
- We analyze when constrained rewriting is useful: gains concentrate on short, underspecified queries, while domain-dense queries are often safely routed to identity through fallback.

2 Related Work

LLM query rewriting for RAG. HyDE generates hypothetical documents for zero-shot retrieval (Gao et al., 2023a); Query2doc uses few-shot prompting to expand queries with pseudo-documents (Wang et al., 2023). Other work compares prompting strategies and reformulation toolkits (Jagerman et al., 2023; Bigdeli et al., 2026b; Ma et al., 2023). These methods often improve average retrieval quality, but average effectiveness can hide unsafe regressions on individual queries. Our focus is not a new rewrite generator, but an external reliability layer for any LLM-mediated query transformation.

Query drift and selective rewriting. Classical pseudo-relevance feedback already exposed the expansion–drift tradeoff (Rocchio, 1971; Lavrenko and Croft, 2001; Abdul-Jaleel et al., 2004). Query-performance prediction estimates when retrieval quality is likely to be poor (Cronen-Townsend et al., 2002; Shtok et al., 2012). Recent work shows that LLM rewriting can hurt complex, ambiguous, or domain-specific queries (Weller et al., 2024; Killingback and Zamani, 2025; Abe et al., 2025). Newer reformulation systems reduce drift through mutual verification, ensemble prompting, process supervision, retrieval-grounded feedback, ranking feedback, or ranker-guided selection (Jia et al., 2024; Dhole et al., 2024; Ye et al., 2025; Bigdeli et al., 2026a; Mao et al., 2024; Venkatesh et al., 2026). These methods improve query expansion or selection policies. Our goal is narrower: add a black-box guardrail around any rewriting method so unsafe rewrites are converted to identity behavior through validation and fallback.

Fusion and variant selection. Retrieval fusion combines ranked lists from multiple systems or query variants, with reciprocal-rank fusion (RRF) a standard rank-only approach (Cormack et al., 2009). RAG-Fusion applies the same idea to LLM-generated query variants for RAG (Rackauckas, 2024). Fusion is complementary to our wrapper: it can preserve original-query evidence by keeping the original run in the fused list, but it requires additional retrieval calls and still depends on generated variants not overwhelming downstream ranking or generation.

Constrained generation and grounding interfaces. Lexical constraints can be enforced during decoding through constrained beam search or neuro-symbolic decoding (Hokamp and Liu, 2017; Lu et al., 2021, 2022). Those approaches are powerful but require token-level control. Our setting needs a narrower property: preserve evidence anchors that the retriever relies on. A prompt-and-verify wrapper targets this property and works with black-box LLMs.

Adaptive grounded generation. Methods such as Self-RAG and FLARE decide when retrieval should be triggered or refreshed during generation (Asai et al., 2024; Jiang et al., 2023b). Our work is complementary: we focus on the query transformation layer that determines which evidence

is available to any downstream generator. In that sense, evidence-anchor preservation is a precondition for answer-level faithfulness rather than an alternative to it.

Faithfulness and attribution in RAG. Evaluation work on grounded generation typically asks whether generated claims are entailed by, attributable to, factually supported by, or correctly cited against retrieved documents (Rashkin et al., 2023; Gao et al., 2023b; Min et al., 2023). AIS evaluates whether generated claims are supported by identified or provided sources; our work targets the earlier question of whether benchmark-relevant sources enter that provided set. Our experiments do not measure final-answer faithfulness directly. Instead, they isolate an upstream condition for such faithfulness: whether LLM-mediated query transformations preserve access to the evidence that answer-level systems would need.

3 Constraint-Based Rewriting

3.1 Problem Setup

Let $q = (t_1, \dots, t_n)$ be an original query, $\mathcal{R}(q)$ an LLM rewriting function producing q' , and $\mathcal{M}(q, \mathcal{D})$ a retrieval metric over corpus $\mathcal{D}$, here nDCG@10. We compute

$$\Delta(q) = \mathcal{M}(q', \mathcal{D}) - \mathcal{M}(q, \mathcal{D})$$

and mark a rewrite as harmful when $\Delta(q) < -0.05$. This is a qrel-observed harm definition: it captures loss under the benchmark relevance labels, not every possible source that might support an answer. The goal is not to maximize rewriting aggressiveness, but to permit expansion only when the rewrite remains compatible with evidence retrieval.

3.2 Evidence-Anchor Constraints

We define three anchor classes and one global edit constraint.

Rare terms. $T_{\text{rare}} = \{t \in q \mid \text{IDF}(t) > 3.5\}$. IDF is computed per corpus as $\ln(N/\text{df}_t)$, where df_t is the document frequency of t after the same analyzer used for BM25 tokenization. Terms above this threshold appear in fewer than roughly 3% of documents and often carry the retrieval signal in specialized corpora. A query term that does not occur in the corpus has $\text{df}_t = 0$, for which $\ln(N/\text{df}_t)$ is undefined; such terms are treated

as rare anchors, which is the conservative direction because it protects a term the retriever cannot match anywhere else. Because the threshold is applied to a per-corpus statistic, the same term can be an anchor in one corpus and not in another.

Named entities. $E = \text{NER}(q)$, preserved with exact span matching. This protects multi-token anchors such as *Back to the Future* or *FEMA*.

Numerics. $\mathcal{N} = \{t \in q \mid t \sim [0-9]^+\}$, preserving quantities, years, and other numeric constraints. The pattern matches maximal digit runs, so an alphanumeric identifier such as *GPT-4* or *7B* is protected as a unit only when the named-entity or rare-term rule also selects it; the numeric rule alone protects the digit run within it. Decimals and ranges are therefore protected component-wise rather than as single spans.

Edit ratio. $\text{EditRatio}(q, q') = \text{EditDist}(q, q') / \max(|q|, |q'|) \leq 0.5$. This rejects complete paraphrases while allowing expansion.

The full constraint is:

$$\mathcal{C}(q, q') = C_{\text{rare}} \wedge C_{\text{entity}} \wedge C_{\text{numeric}} \wedge C_{\text{edit}}.$$

These constraints are intentionally simple: each can be checked deterministically, which makes the wrapper independent of model internals.

Validation and fallback. The validation step uses case-insensitive substring matching for rare terms and entities, regex matching for numerics, and Levenshtein distance for edit ratio. Substring matching does not enforce token boundaries, so a short anchor can be satisfied by a longer word that contains it; this makes validation more permissive than the prompt’s *verbatim* instruction and means the check is a necessary rather than a sufficient condition for true preservation. Informal auditing during development suggested that validation misses concentrate in morphological variants; we did not run a sized annotation study. Validation passes for 40–58% of queries depending on dataset; the rest fall back to identity. This fallback rate is an expected safety behavior: it limits upside, but it is also the mechanism that prevents the rewriter from corrupting already evidence-dense queries. A safety-relevant residual failure is *false acceptance*: the rewrite preserves detected anchors but still hurts retrieval by adding misleading context, changing emphasis, or exploiting an anchor that is not sufficient for relevance.

Original	Unconstrained	Constrained	Δ
is <u>remdesivir</u> effective for COVID-19	antiviral therapies for respiratory infections	<u>remdesivir</u> treatment for COVID-19	+0.02
<u>FEMA</u> flood zone <u>insurance</u> requirements	government disaster protection regulations	<u>FEMA</u> flood zone <u>insurance</u> requirements	0.00
who plays doc in <u>Back to the Future</u>	famous movie character actors	who plays doc in <u>Back to the Future</u> film	+0.08

Table 1: Typical evidence-anchor drift. Unconstrained rewrites remove candidate evidence anchors; constrained rewrites preserve them or fall back. Δ is nDCG@10 change vs. identity.

Algorithm 1 Direct Constrained Query Rewriting

Require: Query q , IDF statistics, rewriting method $\mathcal{R}$

- 1: $A \leftarrow \{t \in q : \text{IDF}(t) > 3.5\} \cup \text{UNER}(q) \cup \{t \in q : t \sim [\emptyset-9]^+\}$
- 2: Prompt $\mathcal{R}$ to rewrite q while preserving all anchors in A
- 3: $q' \leftarrow \mathcal{R}(q, A)$
- 4: **if** $A \subseteq q'$ and $\text{EditRatio}(q, q') \leq 0.5$ **then**
- 5: **return** q'
- 6: **else**
- 7: **return** q {identity fallback}
- 8: **end if**

3.3 Applying the Wrapper

Algorithm 1 is method-agnostic, but the validation target differs for replacement and expansion methods. For direct rewriting, the LLM output replaces the query, so we enforce all anchor constraints and the edit-ratio cap. For the HyDE-style and Query2doc-style conditions, the generated text is an expansion artifact rather than a replacement query; we require anchor preservation in the generated artifact and retain the original query in the final retrieval input. If the expansion artifact drops required anchors or violates the method-specific format, the system falls back to the original query. This distinction prevents the edit-ratio cap from incorrectly rejecting intentionally long pseudo-documents.

4 Experimental Setup

Datasets. We evaluate on four BEIR benchmarks (Thakur et al., 2021) and Natural Questions (Kwiatkowski et al., 2019): NFCorpus (323 queries, medical), SciFact (300, scientific claims), TREC-COVID (50, biomedical), FiQA-2018 (648, finance), and NQ (300 sampled queries, open-domain QA). Table 2 summarizes them.

Dataset	Queries	Corpus	Domain
NFCorpus	323	3.6K	medical
SciFact	300	5K	claims
TREC-COVID	50	171K	biomedical
FiQA	648	57K	finance
NQ	300	2.7M	open QA

Table 2: Evaluation datasets. NQ is a fixed 300-query sample used for general-domain validation.

Retrievers. We use BM25 (Robertson and Zaragoza, 2009), SPLADE-cocondenser-ensembledistil (Formal et al., 2021), and E5-base-v2 with FAISS (Wang et al., 2022; Johnson et al., 2021).

Rewriting methods and models. The main experiments use Zephyr-7B-beta (Tunstall et al., 2023). We compare identity, unconstrained direct rewriting, constrained direct rewriting, HyDE-style expansion, constrained HyDE-style expansion, Query2doc-style expansion, constrained Query2doc-style expansion, and an append-only baseline. Append-only uses the same LLM expansion as direct rewriting but always prepends the original query verbatim, so original lexical evidence remains in the retrieval input. We also validate with Mistral-7B-v0.1 (Jiang et al., 2023a). These two checkpoints are not independent model families: Zephyr-7B-beta is an instruction-tuned derivative of Mistral-7B-v0.1, so the comparison isolates the effect of instruction tuning on rewriting behaviour rather than demonstrating generalization across architectures.

Status of the expansion baselines. We label the expansion conditions *HyDE-style* and *Query2doc-style* because they are adaptations rather than faithful reimplementations of the cited methods. Query2doc as published uses few-shot prompting to generate pseudo-documents, whereas our prompt is zero-shot. HyDE as published is formulated as hypothetical-document embedding for

zero-shot dense retrieval, whereas we use the generated text as an expansion across lexical, learned-sparse, and dense retrievers. These conditions therefore should not be read as reproductions of the published numbers for either method. They also test a different property from the direct-rewriting condition. In the constrained versions of these two conditions the original query is retained in the final retrieval input, as described in Section 3, so the original lexical anchors are present there whether or not the generated artifact repeats them. What the check rejects in those conditions is therefore a generated artifact that drops the anchors, not a retrieval input that lacks them.

Metrics. The primary effectiveness metric is qrel-judged nDCG@10, chosen because top-ranked retrieved documents are the evidence budget exposed to a typical RAG generator. We use the standard BEIR qrels, treat unjudged documents as non-relevant, and macro-average per-query nDCG. The primary grounding-safety metric is qrel-observed harm rate: the fraction of queries with $\Delta \text{nDCG@10} < -0.05$ relative to identity. These metrics measure benchmark-labeled evidence exposure, not generated-answer faithfulness or exhaustive answer support. They are appropriate for our scoped question: whether LLM-mediated query transformations preserve access to evidence that answer-level systems would need, while inheriting the usual incompleteness bias of qrel-based IR evaluation. Outcomes are paired by query, so paired bootstrap for nDCG and McNemar tests for harm incidence are the appropriate analyses; we do not report them here, and Appendix B states what the released material does and does not contain.

Efficiency and reproducibility. In our local serving setup, the constraint wrapper added approximately 153 ms per query with retrieval time excluded; the number should be interpreted as implementation-specific rather than a deployment guarantee. The overhead is dominated by constrained prompting and validation; no retriever-side indexing or training is added. All datasets are public, all main models are open-weight, and Appendix B specifies the prompts and the main configuration settings, and states which implementation details are not reported.

5 Results

5.1 Main Results

Table 3 shows BM25 results. Unconstrained rewriting is the worst-performing condition on all four BEIR datasets, dropping macro nDCG@10 from 0.438 to 0.208. Constrained rewriting recovers near-identity mean performance (0.437) while reducing mean harm from 48% to 10%. The method does not merely preserve all tokens: append-only rewriting reduces harm, but the targeted constraint layer yields lower harm on every BEIR dataset.

The TREC-COVID result is instructive. It has the highest unconstrained harm (70%) and the highest residual constrained harm (20%), suggesting that specialized biomedical queries remain difficult even when anchor preservation is enforced. Two explanations are consistent with the guardrail’s design and we did not run the analysis that separates them. Under *false acceptance*, the detected anchors survive and validation passes, but the accepted rewrite still hurts retrieval by adding misleading context or shifting emphasis away from the aspect the qrels reward. Under *undetected anchors*, the term that carried the evidence was never selected as an anchor, so preservation was never required of it; this is plausible in a corpus where much of the discriminative vocabulary is domain-common and therefore below the IDF threshold. Distinguishing the two requires the anchor-level analysis described in the limitations, and the distinction matters operationally: false acceptance argues for a stricter acceptance rule, whereas undetected anchors argue for a lower threshold or a domain term list. We also note that 20% of TREC-COVID’s 50 queries is 10 queries, so this residual is estimated from a small sample and carries wide uncertainty.

5.2 Retriever Generalization

Table 4 tests whether semantic retrievers compensate for anchor loss. They do not. Dense E5 suffers unconstrained harm rates comparable to BM25, including 76% on TREC-COVID. Across 12 dataset–retriever settings, constrained rewriting reduces harm from 31–76% to 2–20%.

5.3 Ablations and Worst-Case Behavior

Table 5 ablates constraint types on NFCorpus. Each single-class configuration recovers most of the effect, entity-only being the weakest, and the

Method	NFCorpus	SciFact	TREC-COVID	FiQA	Macro
Original	0.306	0.652	0.577	0.217	0.438
Unconstrained	0.126 (50)	0.358 (42)	0.242 (70)	0.104 (31)	0.208
HyDE-style	0.276 (36)	0.635 (25)	0.530 (52)	0.165 (30)	0.402
Query2doc-style	0.319 (21)	0.620 (20)	0.558 (46)	0.162 (28)	0.415
Append-only	0.298 (15)	0.644 (18)	0.561 (38)	0.189 (22)	0.423
Constr. HyDE-style	0.301 (8)	0.649 (7)	0.570 (24)	0.195 (16)	0.429
Constr. Q2D-style	0.314 (6)	0.647 (6)	0.572 (22)	0.192 (15)	0.431
Constrained	0.307 (5)	0.658 (3)	0.575 (20)	0.207 (11)	0.437

Table 3: BM25 nDCG@10 with harm rate in parentheses. Bold nDCG marks the best rewriting method; bold harm marks the lowest harm rate.

Retriever	Method	NFCorpus	SciFact	TREC-COVID	FiQA
BM25	Original	0.306	0.652	0.577	0.217
	Unconstrained	0.126 (50)	0.358 (42)	0.242 (70)	0.104 (31)
	Constrained	0.307 (5)	0.658 (3)	0.575 (20)	0.207 (11)
SPLADE	Original	0.353	0.708	0.725	0.356
	Unconstrained	0.152 (50)	0.404 (44)	0.354 (66)	0.195 (32)
	Constrained	0.348 (7)	0.710 (2)	0.727 (14)	0.341 (13)
Dense E5	Original	0.354	0.719	0.696	0.399
	Unconstrained	0.150 (54)	0.406 (46)	0.284 (76)	0.221 (36)
	Constrained	0.347 (6)	0.717 (4)	0.683 (14)	0.380 (16)

Table 4: nDCG@10 across retrievers, with harm rate in parentheses.

Configuration	nDCG@10	Harm
No constraints	0.126	50%
Rare only	0.305	6%
Entity only	0.296	8%
Numeric only	0.304	5%
Full constraints	0.307	5%

Table 5: Constraint ablation on NFCorpus with BM25.

full configuration is best on nDCG while tying the numeric-only configuration on harm. The differences among the constrained rows are small and we do not test them for significance.

Worst-case behavior improves as well. The 5th-percentile Δ nDCG@10 for unconstrained rewriting reaches -1.000 on SciFact and -0.910 on TREC-COVID. Under constrained rewriting, the corresponding values are 0.000 and -0.170 ; across BEIR, worst-case damage is reduced by 69–100%.

5.4 Coverage and What the Harm Reduction Includes

The harm reduction reported above is not produced by better rewriting alone. Validation passes for 40–58% of queries depending on dataset, so 42–60% of queries are routed to identity. A fallback query has $\Delta = 0$ by construction and there-

fore cannot be counted as harmful under the $\Delta < -0.05$ criterion. The aggregate harm rate consequently combines two distinct effects: a reduction in the harm caused by rewrites that are accepted, and the removal of a large fraction of queries from any risk of harm at all through abstention. Both are operationally useful, but they are different claims, and the aggregate number does not separate them.

The one acceptance-conditional measurement we report is on Natural Questions, where accepted constrained rewrites have mean Δ nDCG@10 of $+0.014$ against $+0.007$ over all queries (Table 6); since fallback contributes exactly zero, the gap is consistent with roughly half of NQ queries being accepted. We do not report acceptance-conditional harm for every dataset, retriever, and model, and we do not report risk–coverage curves obtained by varying the IDF and edit thresholds.

The comparison this leaves open is the important one. Because abstention alone lowers harm, the relevant question is whether this validator gives a better risk–coverage tradeoff than a simpler rule that abstains on the same fraction of queries, such as random rejection, length-based rejection, or embedding-similarity rejection matched to the same acceptance rate. We did not run those matched-coverage baselines. Our results therefore support the claim that verify-then-fallback is safer

Method	nDCG@10	Harm	vs. id.
Original	0.233	–	–
Unconstrained	0.160	26%	-0.073
Constrained	0.240	6%	+0.007

Table 6: Natural Questions BM25 results on 300 sampled queries. Accepted constrained rewrites have mean Δ nDCG@10 of +0.014.

than unconstrained rewriting; they do not establish that anchor-based validation is the specific reason, and readers should treat the mechanism as motivated by the qualitative examples and the ablation in Table 5 rather than as isolated by a controlled comparison.

5.5 When Does Rewriting Help?

Constrained rewriting is most useful when the original query is underspecified. Across BEIR, short queries (≤ 6 tokens) gain +0.012 mean nDCG@10 with 4% harm and 38% fallback. Long queries (> 12 tokens) show -0.004 mean change with 12% harm and 68% fallback. Queries with high rare-term ratio also fall back more often (62%), consistent with the intended behavior: preserve identity when the query is already evidence-dense.

On Natural Questions, constrained rewriting slightly exceeds identity (0.240 vs. 0.233 nDCG@10) while reducing harm from 26% to 6% (Table 6). This supports the same pattern in a general-domain setting: expansion helps when queries are short, but must be prevented from deleting evidence anchors.

5.6 Checkpoint Comparison

Table 7 compares the two checkpoints. Zephyr’s unconstrained rewrites produce higher harm rates on all four datasets, and constrained harm is lower than unconstrained harm for both checkpoints. We recorded harm rates only for this sweep, so rewritten-query nDCG@10, mean deltas, and acceptance rates are not available for Mistral and we make no claim about their relative fallback rates. The two checkpoints also share a base model, so this is evidence that the guardrail survives the removal of instruction tuning, not evidence of generalization across model families. Mistral-7B-v0.1 is a pretrained base checkpoint rather than an instruction-tuned chat model, so the statement that prompts use a default chat template does not describe it, and we do not report the exact serializa-

Model	Data	Id.	Unc.	Con.
Zephyr	NFCorpus	0.306	50%	5%
Mistral	NFCorpus	0.306	25%	3%
Zephyr	SciFact	0.652	42%	3%
Mistral	SciFact	0.652	13%	3%
Zephyr	TREC-COVID	0.577	70%	20%
Mistral	TREC-COVID	0.577	34%	6%
Zephyr	FiQA	0.217	31%	11%
Mistral	FiQA	0.217	16%	4%

Table 7: Checkpoint comparison with BM25. Id. is identity nDCG@10; Unc. and Con. are harm rates, not nDCG. Only harm rates were recorded for this sweep; rewritten-query nDCG, mean deltas, and acceptance rates are not available.

tion used for that checkpoint.

6 Discussion

Grounding fails upstream of generation. RAG faithfulness is often evaluated at answer time, but our results show that an LLM can damage grounding at the query interface. Once evidence anchors are removed, even a faithful generator may answer from the wrong retrieved context. Query rewriting should therefore also be evaluated with grounding-safety metrics, not only mean retrieval effectiveness.

What retrieval-side evaluation shows. Generation-side faithfulness would answer a different question: whether a particular answerer uses retrieved evidence faithfully. Our question is earlier in the pipeline: whether rewriting preserves access to task-relevant evidence in the retrieved context. Such evidence access is not sufficient for faithful answering, but it is an operational prerequisite for citation, attribution, and evidence-grounded reasoning in non-oracle retrieval-grounded systems. A rewrite that removes the relevant source has already failed as a grounding interface, even if some generators later mask the failure through parametric knowledge.

Why not always use identity? Identity is a strong baseline, especially for domain-dense queries. But universal abstention gives up real gains on short and underspecified queries, as seen on Natural Questions and the short-query stratum. The practical design target is controlled expansion: permit the LLM to add context, but require it to preserve the lexical evidence needed by retrieval.

Relation to fusion baselines. Original-plus-rewrite fusion is a natural alternative because it

can retain the original query as an independent retrieval signal. We include append-only rewriting as a single-call conservative baseline, but RRF over original and rewritten runs remains the most important missing safety baseline. The wrapper should be viewed as a pre-retrieval validation layer that may be combined with fusion when multiple retrieval calls are affordable.

Scope of the contribution. The contribution is not a new retriever and not a claim that rewriting should always improve over identity. It is a risk-reducing interface for systems that already use LLM-mediated query transformation. In that setting, the relevant comparison is not only whether rewriting beats identity on average, but whether a rewriting component can be made less likely to remove evidence needed downstream.

7 Conclusion

LLM query rewriting can damage retrieval-side grounding before generation begins, and harmful rewrites in our data frequently drop the rare terms, entities, and numerics that retrieval depends on. Across datasets, retrievers, rewriting methods, and checkpoints, this failure is common for unconstrained rewriting and persists even with dense retrieval. A simple constraint layer—identify candidate anchors, prompt for preservation, verify deterministically, and fall back to identity—reduces qrel-observed evidence-exposure harm while preserving near-identity retrieval effectiveness and allowing gains in short-query strata. That reduction combines safer accepted rewrites with abstention on 42 to 60% of queries, and we do not separate the two. The result is not an end-to-end guarantee of faithful generation; it is a practical guardrail for preserving evidence that downstream grounded-language-model systems often need to produce attributable, evidence-supported answers.

Limitations

The mechanism is not isolated. The guarded conditions in this paper combine anchor preservation with identity fallback, and the direct-rewriting conditions add the edit-ratio cap; the expansion conditions omit the cap so that long pseudo-documents are not rejected for length alone. We do not report an edit-ratio-only condition, an anchor-only condition without the edit cap, a constrained prompt without validation or fallback, or post-hoc

anchor validation applied to unconstrained outputs. The ablation in Table 5 varies which anchor classes are required but retains the edit cap in every constrained row, and the numeric-only row reaches 0.304 nDCG and 5% harm, within 0.003 nDCG of the full configuration; we did not measure how many NFCorpus queries contain numerics, so we cannot rule out that the shared cap and fallback rather than the numeric rule account for most of that row. Our results therefore support the weaker statement that selective rewriting with deterministic validation and fallback is safer than unconstrained rewriting. They do not establish that preserving evidence anchors is the operative cause, which is what the title names. Readers should treat anchor preservation as the motivating hypothesis, supported by the qualitative pattern in Table 1 and by the relative ordering of the anchor-class rows, rather than as an isolated and measured cause.

Harm reduction is confounded with coverage.

As set out in Section 5.4, 42–60% of queries are routed to identity and cannot register as harmful. We do not report acceptance-conditional harm for every setting, risk-coverage curves over the IDF and edit thresholds, or matched-coverage abstention baselines such as random, length-based, or similarity-based rejection at the same acceptance rate. Without those, we cannot show that anchor-based validation beats a simpler abstention rule of equal coverage.

The expansion comparison is not clean. We do not specify whether the unconstrained HyDE-style and Query2doc-style baselines retain the original query in the final retrieval input, whereas the constrained versions of those conditions do. If they do not, the constrained expansion rows differ from their unconstrained counterparts in both anchor prompting and query retention, and the append-only row is the more direct isolation of query retention.

Statistical support is limited. We do not report confidence intervals on the harm proportions, and an interval on a single proportion would not in any case test whether two methods differ on the same queries. We do not report paired bootstrap or randomization intervals for nDCG deltas, and we do not report McNemar or paired exact tests for harm incidence. Several central differences are small enough that this matters: identity against constrained BM25 macro nDCG is 0.438 against

0.437, identity against constrained NQ nDCG is 0.233 against 0.240, and append-only against constrained macro nDCG is 0.423 against 0.437. We do not claim these differences are statistically distinguishable, and no multiple-comparison correction is applied across the many dataset and retriever cells.

The harm threshold has a floor effect. Harm is defined as $\Delta \text{nDCG}@10 < -0.05$. Because nDCG is bounded below by zero, a query whose identity nDCG@10 is at or below 0.05 has little or no room to be classified as harmful. That floor differs across datasets and retrievers, and it is most restrictive exactly where identity scores are lowest, such as FiQA. We do not report harm at several thresholds or conditional on the identity score, so cross-dataset harm rates should be compared with this in mind.

nDCG@10 is an indirect measure of evidence access. Our conceptual claim concerns whether labeled evidence remains reachable, but nDCG@10 is a ranking-quality measure. It can fall while the same relevant documents remain in the top ten, and evidence can be lost while graded gain moves only modestly. We do not report Recall@10 or Recall@ k , hit rate for at least one qrel-positive document, the number of qrel-positive documents in the retrieved context, the rank of the first relevant document, or the overlap between the original and rewritten relevant sets, each of which would speak more directly to evidence exposure.

Anchor deletion is not shown to explain the harmful rewrites. We give illustrative examples but no systematic anchor-level analysis. We do not report the proportion of rewrites that delete each anchor class, harm conditional on deleting a rare term, entity, or numeric, harm when all detected anchors are retained, anchor-detector precision or recall, false-acceptance and false-rejection rates, or the size and annotation procedure of the manual audit referenced in Section 3. The paper therefore identifies a plausible mechanism without demonstrating the fraction of measured regressions it accounts for.

Model coverage is narrow. Zephyr-7B-beta is an instruction-tuned derivative of Mistral-7B-v0.1, so the two checkpoints share a base and are not independent model families. The Mistral sweep records harm rates only. We evaluate English queries and no frontier or closed model.

Threshold selection and cost. The IDF threshold of 3.5 and the edit-ratio cap of 0.5 were fixed across corpora, and the reported sensitivity check was run on SciFact. We do not report a held-out threshold-selection protocol, so we cannot rule out that these values reflect behaviour observed on evaluation data. The 153 ms wrapper overhead was measured on a single local serving setup with retrieval excluded; without hardware, serving stack, precision, batching, and output-length statistics it should be read as an order of magnitude rather than a portable figure.

Scope already stated. We do not claim that anchor preservation guarantees factual or faithful answers. Our experiments measure qrel-observed retrieval-side evidence exposure, and a generation-side evaluation could reveal cases where retrieval harm is masked by parametric knowledge, or where high nDCG still leads to unsupported generated claims. The anchor detector is deliberately simple and can be brittle under aliases, morphology, spelling variation, multilingual queries, or NER errors, and case-insensitive substring validation does not enforce token boundaries. IDF thresholds depend on corpus statistics that may be unavailable in closed retrieval systems. TREC-COVID has only 50 queries, so its estimates carry wide uncertainty. We also do not evaluate retrieval fusion baselines such as reciprocal-rank fusion over original and rewritten queries (Cormack et al., 2009), which may further reduce risk at higher retrieval cost.

Ethics Statement

This work studies retrieval robustness using public benchmarks and open-weight models. The main risk is over-trusting a guardrail as a guarantee. We therefore report residual harm and recommend deployment with task-specific validation, especially in high-stakes domains.

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A Additional Experimental Details

All reported constrained methods use the same anchor extraction and validation procedure. The IDF threshold is fixed at 3.5 across corpora. Threshold sensitivity on SciFact shows a stable plateau from 3.0 to 4.5; below 2.5 too many common terms are protected, and above 4.5 too few terms are protected. Validation uses substring matching for preserved terms and entities, regex matching for numerics, and Levenshtein distance for edit ratio.

B Reproducibility Checklist

Prompts. Prompts use the checkpoint’s default chat template with temperature 0 for Zephyr-7B-beta. Mistral-7B-v0.1 is a pretrained base checkpoint and has no instruction-tuned chat template; we do not report the serialization used for it. The

unconstrained direct prompt is: “Rewrite the query to improve document retrieval. Preserve the user’s intent and return only the rewritten query.” The constrained direct prompt adds: “The following evidence anchors must appear verbatim in your output: {anchors}. Do not remove or paraphrase them.” The HyDE-style prompt is: “Write one concise hypothetical passage that would answer the query and contain terms useful for retrieval. Return only the passage.” Its constrained version adds the same verbatim-anchor sentence. The Query2doc-style prompt is: “Generate a short pseudo-document containing likely relevant terms for retrieving documents about the query. Return only the pseudo-document.” Its constrained version adds the same verbatim-anchor sentence. Append-only uses the direct rewrite prompt and forms the retrieval input as `original query || rewrite`.

Implementation configuration. Rewriting uses greedy decoding with a 100-token cap for direct rewrites and append-only expansions, and a 256-token cap for HyDE and Query2doc expansions. Anchor extraction lowercases and tokenizes queries before IDF lookup; document frequencies are computed over the retrieval corpus after the same analyzer used for BM25 tokenization. Validation uses case-insensitive substring matching for terms and entities, regex matching for numerics, and token-level Levenshtein distance for edit ratio. Entity spans are produced by an off-the-shelf English NER component applied to the original query. We do not report the exact package, model revisions, or library versions, which is a reproducibility gap we note rather than paper over. Latency was measured on the same local serving setup used for rewriting, with retrieval time excluded from the reported 153 ms wrapper overhead.

Retrieval configuration. BM25 uses Okapi BM25 with $k_1 = 1.5$ and $b = 0.75$. SPLADE uses SPLADE-cocondenser-ensembledistil. Dense retrieval uses E5-base-v2 embeddings with FAISS; E5 query and document prefixes follow the model release. We do not report the FAISS index type and similarity function, the embedding pooling and normalization, the input truncation policy, or the hardware used, each of which would be needed for exact replication of the dense-retrieval numbers. The NQ validation uses the same fixed 300-query sample for all conditions.

Artifact release. The supplement published with the submission documents the prompt templates, configuration fields, and the schema for a per-query release covering query identifiers, model outputs, detected anchors, validation decisions, fallback indicators, per-query nDCG@10 and deltas, and retrieved document identifiers. That populated release is not included with this version. We state plainly that the central claim of this paper is a paired, query-level comparison, and that a reader cannot audit it from the aggregate tables alone.